



COMPUTER
ENGINEERING

QUEZON CITY

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ACTIVITY NO. 10 Install, Configure and Manage Enterprise Log Monitoring via Ansible

1. PROCEDURE OUTPUT

For this hands-on activity, we will be installing the ELK stack which consists of Elasticsearch, Kibana, and Logstash. However, a few constraints are to be accounted for. Security is disabled for simplicity but should not be done so in actual production to avoid exposure to the internet or untrusted networks. The three components are split into three different virtual machines due to resource constraints. The minimum requirements of 8 GB per node are unattainable with my current system specifications (4 GB for workstation, 2 GB for remote hosts).

ansible.cfg

```
GNU nano 7.2 ansible.cfg
[defaults]
inventory = inventory.yaml
host_key_checking = False
```

This is the config file to set the used inventory.yaml file to the one inside the current directory. Host_key_checking disables SSH host-key verification prompts.

inventory.yaml

```
all:
  children:
    elasticsearch_servers:
      hosts:
        ubuntu-workstation:
          ansible_host: 192.168.56.105
          ansible_connection: local
    kibana_servers:
      hosts:
        ubuntu-node:
          ansible_host: 192.168.56.104
    logstash_servers:
      hosts:
        centos-node:
          ansible_host: 192.168.56.103
```

For the inventory.yaml file, the servers are sorted by elasticsearch_servers, kibana_servers, and logstash_servers depending on what is needed to be installed. The workstation is assigned to Elasticsearch, the Ubuntu remote host is assigned to Kibana, and the CentOS remote host is assigned to Logstash.

site.yaml

```
mdmlot@workstation: ~/site
GNU nano 7.2
--
- name: Apply updates to all managed nodes
  hosts: all
  become: true
  roles:
    - common

- name: Install and configure Elasticsearch on Ubuntu workstation
  hosts: elasticsearch_servers
  become: true
  roles:
    - elasticsearch

- name: Install and configure Kibana on Ubuntu remote host
  hosts: kibana_servers
  become: true
  roles:
    - kibana

- name: Install and configure Logstash on CentOS remote host
  hosts: logstash_servers
  become: true
  roles:
    - logstash
```

3 roles are created, named according to the required installation. Common is used for applying updates to all the managed nodes.

roles/elasticsearch/tasks/main.yml

```
# tasks file for elasticsearch for ubuntu
- name: install apt prerequisites
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - gnupg
      - ca-certificates
    state: present
    update_cache: true
```

Installs apt-transport-https, gnupg, and ca-certificates. These are needed because Elastic's repository is served over HTTPS and signed with a GPG key not present on base Ubuntu installations.

```
- name: download Elastic GPG key
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /usr/share/keyrings/elastic.asc
    mode: '0644'
```

Retrieves Elastic's signing key into /usr/share/keyrings/. Without this, apt would refuse to trust packages coming from Elastic's repo so the key has to be on disk before the repo is added.

```
- name: add Elastic apt repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elastic.asc] https://artifacts.elastic.co/packages/9.x/apt stable main"
    filename: elastic-9.x
    state: present
    update_cache: true
```

Registers the 9.x Elastic repo, explicitly bound to the key we just downloaded via signed-by=. This is what makes apt install elasticsearch pull the correct version instead of an outdated distro package.

```
- name: set vm.max_map_count required by Elasticsearch
  ansible.posix.sysctl:
    name: vm.max_map_count
    value: '1048576'
    sysctl_set: true
    state: present
    reload: true
```

Raises the kernel's vm.max_map_count to 1048576 for the memory-mapped index files. Without this high limit, the JVM will refuse to start with an obscure bootstrap error.

```
- name: install Elasticsearch
  ansible.builtin.apt:
    name: elasticsearch
    state: present
```

Installs the package now that the repo, key, and kernel setting are in place.

```
- name: set Elasticsearch heap size to 1GB
  ansible.builtin.copy:
    dest: /etc/elasticsearch/jvm.options.d/heap.options
    content: |
      -Xms1g
      -Xmx1g
    owner: root
    group: elasticsearch
    mode: '0660'
```

Writes a JVM options drop-in capping the heap at 1 GB to keep the host responsive.

```
- name: create Elasticsearch configuration file
  ansible.builtin.copy:
    dest: /etc/elasticsearch/elasticsearch.yml
    content: |
      cluster.name: hoa10-cluster
      node.name: ubuntu-workstation
      path.data: /var/lib/elasticsearch
      path.logs: /var/log/elasticsearch
      network.host: 0.0.0.0
      http.port: 9200
      discovery.type: single-node
      xpack.security.enabled: false
      xpack.security.enrollment.enabled: false
      xpack.security.transport.ssl.enabled: false
      xpack.security.http.ssl.enabled: false #all disabled for lab purposes only
    owner: root
    group: elasticsearch
    mode: '0660'
```

Writes elasticsearch.yml with the cluster name, node name, network binding, and X-Pack disabled. Disabling X-Pack avoids the postinstall SSL/password mismatch that prevents the service from starting (Constraint: Security Risk for the sake of simplicity and demonstration).

```
- name: Clear stale X-Pack SSL entries from Elasticsearch keystore
  ansible.builtin.shell: |
    /usr/share/elasticsearch/bin/elasticsearch-keystore remove xpack.security.transport.ssl.keystore.secure_password || true
    /usr/share/elasticsearch/bin/elasticsearch-keystore remove xpack.security.transport.ssl.truststore.secure_password || true
    /usr/share/elasticsearch/bin/elasticsearch-keystore remove xpack.security.http.ssl.keystore.secure_password || true
    /usr/share/elasticsearch/bin/elasticsearch-keystore remove xpack.security.http.ssl.truststore.secure_password || true
  become: true
  become_user: elasticsearch
```

Removes the auto-generated SSL passwords that Elasticsearch's postinstall script writes into its secure keystore when the package is first installed.

```
- name: Start and enable Elasticsearch service
  ansible.builtin.systemd:
    name: elasticsearch
    state: restarted
    enabled: true
    daemon_reload: true
```

Enables the unit at boot and restarts it now so the freshly written config is picked up. daemon_reload is included so systemd re-reads any unit changes from the package install.

```
- name: wait for Elasticsearch to answer on port 9200
  ansible.builtin.uri:
    url: http://127.0.0.1:9200
    status_code: 200
  register: es_up
  until: es_up.status == 200
  retries: 30
  delay: 5
```

Polls the REST API until it returns 200. Elasticsearch takes 10–30 seconds to bootstrap, and without this wait, later plays would race against a still-starting Elasticsearch and fail.

roles/kibana/tasks/main.yml

```
# tasks file for kibana for ubuntu remote
- name: install apt prerequisites
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - gnupg
      - ca-certificates
    state: present
    update_cache: true
```

HTTPS and GPG tooling aren't installed by default, and Kibana's repo needs both.

```
- name: download Elastic GPG key
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /usr/share/keyrings/elastic.asc
    mode: '0644'
```

Places the signing key where apt can validate packages from the Elastic repo.

```
- name: add Elastic apt repository
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/usr/share/keyrings/elastic.asc] https://artifacts.elastic.co/packages/9.x/apt stable main"
    filename: elastic-9.x
    state: present
    update_cache: true
```

Registers the 9.x repo pointing at the same key, so Kibana matches the Elasticsearch version already running.

```
- name: install Kibana
  ansible.builtin.apt:
    name: kibana
    state: present
```

Installs Kibana from the verified repo.

```
- name: limit Kibana memory use to 768MB
  ansible.builtin.lineinfile:
    path: /etc/kibana/node.options
    regexp: '^--max-old-space-size='
    line: "--max-old-space-size=768"
```

Sets Node.js's max old-space size to 768 MB because Kibana's default memory profile assumes a much beefier host, and without this cap the Ubuntu remote host (2 GB RAM) would swap heavily.

```
- name: create Kibana configuration file
ansible.builtin.copy:
  dest: /etc/kibana/kibana.yml
  content: |
    server.port: 5601
    server.host: "0.0.0.0"
    server.name: "ubuntu-node"
    elasticsearch.hosts: ["http://192.168.56.105:9200"]
  owner: root
  group: kibana
  mode: '0660'
```

Writes kibana.yml with the port, host binding, and the URL of the Elasticsearch node. Kibana needs to know exactly where ES lives since it's on a different machine in this topology.

```
- name: Start and enable Kibana service
ansible.builtin.systemd:
  name: kibana
  state: restarted
  enabled: true
  daemon_reload: true
```

Enables at boot and restarts now so the new config and memory limit take effect immediately.

```
- name: wait for Kibana to listen on port 5601
ansible.builtin.wait_for:
  port: 5601
  host: 127.0.0.1
  timeout: 300
```

Kibana can take 1–2 minutes to come up on a low-RAM host, so we block until the port is listening to guarantee the next steps aren't racing a half-started UI.

roles/logstash/tasks/main.yml

```
- name: import Elastic GPG key
ansible.builtin.rpm_key:
  key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
  state: present
```

Imports Elastic's key into the RPM keyring. On RHEL systems, dnf verifies every package's signature, so the key must be present or the install will fail.

```
- name: add Elastic yum repository
  ansible.builtin.yum_repository:
    name: elastic-9.x
    description: Elastic repository for 9.x packages
    baseurl: https://artifacts.elastic.co/packages/9.x/yum
    gpgcheck: true
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: true
    state: present
```

Adds the 9.x yum repo so dnf install logstash pulls the matching version instead of whatever CentOS ships.

```
- name: install Logstash
  ansible.builtin.dnf:
    name: logstash
    state: present
```

Installs Logstash from the verified repo.

```
- name: set Logstash heap size (min) to 512MB
  ansible.builtin.lineinfile:
    path: /etc/logstash/jvm.options
    regexp: '^Xms'
    line: "-Xms512m"
```

Locks the JVM's initial heap to 512 MB to avoid the JVM's default slow ramp-up on a memory-constrained VM and gives predictable memory use from boot.

```
- name: set Logstash heap size (max) to 512MB
  ansible.builtin.lineinfile:
    path: /etc/logstash/jvm.options
    regexp: '^Xmx'
    line: "-Xmx512m"
```

Caps the JVM max heap at 512 MB. The default is 1 GB on 2 GB hosts which would leave too little room for the OS and Logstash's own pipeline workers on the CentOS node.

```
- name: create Logstash pipeline file
ansible.builtin.copy:
  dest: /etc/logstash/conf.d/syslog-pipeline.conf
  content: |
    input {
      syslog {
        port => 5140
      }
      beats {
        port => 5044
      }
    }

    output {
      elasticsearch {
        hosts => ["http://192.168.56.105:9200"]
        data_stream => false
        index => "syslog-%{+YYYY.MM.dd}"
      }
    }
  owner: root
  group: logstash
  mode: '0640'
```

Writes a pipeline that listens for syslog on 5140 and beats on 5044, then ships events to Elasticsearch under syslog-YYYY.MM.dd. Without this file, Logstash would receive nothing and Elasticsearch would stay empty.

```
- name: open Logstash ports in firewall
ansible.posix.firewalld:
  port: "{{ item }}"
  permanent: true
  immediate: true
  state: enabled
loop:
  - 5140/tcp
  - 5140/udp
  - 5044/tcp
  - 9600/tcp
ignore_errors: true
```

Opens 5140/tcp, 5140/udp, 5044/tcp, and 9600/tcp blocked by default.

```
- name: Start and enable Logstash service
ansible.builtin.systemd:
  name: logstash
  state: restarted
  enabled: true
  daemon_reload: true
```

Enables at boot and restarts now so the pipeline config and heap settings load which actually activates the new configuration.

```
- name: wait for Logstash syslog input on port 5140
ansible.builtin.wait_for:
  port: 5140
  host: 127.0.0.1
  timeout: 300
```

Blocks until Logstash is accepting syslog connections.

Playbook Completion Output

```
mdmlot@workstation:~/CPE212_loterte/hoa10$ ansible-playbook site.yaml --ask-become-pass
BECOME password:

PLAY [Apply updates to all managed nodes] *****

TASK [Gathering Facts] *****
ok: [ubuntu-node]
ok: [ubuntu-workstation]
ok: [centos-node]

TASK [common : Install updates (CentOS)] *****
skipping: [ubuntu-workstation]
skipping: [ubuntu-node]
ok: [centos-node]

TASK [common : Install updates (Ubuntu)] *****
skipping: [centos-node]
changed: [ubuntu-workstation]
changed: [ubuntu-node]
```

```
PLAY [Install and configure Elasticsearch on Ubuntu workstation] *****

TASK [Gathering Facts] *****
ok: [ubuntu-workstation]

TASK [elasticsearch : install apt prerequisites] *****
ok: [ubuntu-workstation]

TASK [elasticsearch : download Elastic GPG key] *****
ok: [ubuntu-workstation]

TASK [elasticsearch : add Elastic apt repository] *****
ok: [ubuntu-workstation]

TASK [elasticsearch : set vm.max_map_count required by Elasticsearch] **
ok: [ubuntu-workstation]

TASK [elasticsearch : install Elasticsearch] *****
ok: [ubuntu-workstation]

TASK [elasticsearch : set Elasticsearch heap size to 1GB] *****
ok: [ubuntu-workstation]

TASK [elasticsearch : create Elasticsearch configuration file] *****
changed: [ubuntu-workstation]
```

```
PLAY [Install and configure Kibana on Ubuntu remote host] *****

TASK [Gathering Facts] *****
ok: [ubuntu-node]

TASK [kibana : install apt prerequisites] *****
changed: [ubuntu-node]

TASK [kibana : download Elastic GPG key] *****
changed: [ubuntu-node]

TASK [kibana : add Elastic apt repository] *****
changed: [ubuntu-node]

TASK [kibana : install Kibana] *****
changed: [ubuntu-node]

TASK [kibana : limit Kibana memory use to 768MB] *****
changed: [ubuntu-node]

TASK [kibana : create Kibana configuration file] *****
changed: [ubuntu-node]

TASK [kibana : Start and enable Kibana service] *****
changed: [ubuntu-node]

TASK [kibana : wait for Kibana to listen on port 5601] *****
ok: [ubuntu-node]
```

```
PLAY [Install and configure Logstash on CentOS remote host] *****

TASK [Gathering Facts] *****
ok: [centos-node]

TASK [logstash : import Elastic GPG key] *****
changed: [centos-node]

TASK [logstash : add Elastic yum repository] *****
changed: [centos-node]

TASK [logstash : install Logstash] *****
changed: [centos-node]

TASK [logstash : set Logstash heap size (min) to 512MB] *****
changed: [centos-node]

TASK [logstash : set Logstash heap size (max) to 512MB] *****
changed: [centos-node]

TASK [logstash : create Logstash pipeline file] *****
changed: [centos-node]

TASK [logstash : open Logstash ports in firewall] *****
changed: [centos-node] => (item=5140/tcp)
changed: [centos-node] => (item=5140/udp)
changed: [centos-node] => (item=5044/tcp)
changed: [centos-node] => (item=9600/tcp)

TASK [logstash : Start and enable Logstash service] *****
changed: [centos-node]

TASK [logstash : wait for Logstash syslog input on port 5140] *****
ok: [centos-node]
```

```
PLAY RECAP *****
centos-node      : ok=10   changed=8    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu-node     : ok=9    changed=7    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
ubuntu-workstation : ok=11   changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
mdm1ot@workstation:~/CPE212_loterte/hoa10$ curl http://192.168.56.105:9200
{
  "name" : "ubuntu-workstation",
  "cluster_name" : "hoa10-cluster",
  "cluster_uuid" : "BtgwdKPtR1m5Gybz5N3giA",
  "version" : {
    "number" : "9.5.5",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "48acba663b09e70b115dd879329141506c377890",
    "build_date" : "2026-09-30T22:12:03.984586187Z",
    "build_snapshot" : false,
    "lucene_version" : "10.5.1",
    "minimum_wire_compatibility_version" : "8.19.0",
    "minimum_index_compatibility_version" : "8.0.0"
  },
  "tagline" : "You Know, for Search"
}
mdm1ot@workstation:~/CPE212_loterte/hoa10$ curl http://192.168.56.105:9200/_cluster/health?pretty
{
  "cluster_name" : "hoa10-cluster",
  "status" : "green",
  "timed_out" : false,
  "number_of_nodes" : 1,
  "number_of_data_nodes" : 1,
  "active_primary_shards" : 49,
  "active_shards" : 49,
  "relocating_shards" : 0,
  "initializing_shards" : 0,
  "unassigned_shards" : 0,
  "unassigned_primary_shards" : 0,
  "delayed_unassigned_shards" : 0,
  "number_of_pending_tasks" : 0,
  "number_of_in_flight_fetch" : 0,
  "task_max_waiting_in_queue_millis" : 0,
  "active_shards_percent_as_number" : 100.0
}

mdm1ot@workstation:~/CPE212_loterte/hoa10$ curl http://192.168.56.104:5601/api/status
{"status":{"overall":{"level":"available"}}}mdm1ot@workstation:~/CPE212_loterte/hoa10$ curl http://192.168.56.103:9600
{"host":"server3","version":"9.5.5","http_address":"0.0.0.0:9600","id":"2500f25b-f87b-430b-ad09-d2337ba181ff","name":"server3","ephemeral_id":"62e15dda-4500-451f-a5e6-67b4ce5126e1","snapshot":null,"status":"green","pipeline":{"workers":2,"batch_size":125,"batch_delay":50}}mdm1ot@workstation:~/CPE212_loterte/hoa10$
```

Website Output

Elasticsearch

http://192.168.56.105:9200

JSONRaw DataHeaders

SaveCopyCollapse AllExpand AllFilter JSON

```
name: "ubuntu-workstation"
cluster_name: "hoa10-cluster"
cluster_uuid: "BtgwDKPtR1m5Gybz5N3giA"
version:
  number: "9.5.5"
  build_flavor: "default"
  build_type: "deb"
  build_hash: "48acba663b09e70b115dd879329141506c377890"
  build_date: "2026-09-30T22:12:03.984586187Z"
  build_snapshot: false
  lucene_version: "10.5.1"
  minimum_wire_compatibility_version: "8.19.0"
  minimum_index_compatibility_version: "8.0.0"
tagline: "You Know, for Search"
```

Kibana

Not Secure http://192.168.56.104:5601/app/home#/

elasticFind apps, content, and more.

Home

Welcome home



Elasticsearch

Create search experiences with a refined set of APIs and tools.



Observability

Consolidate your logs, metrics, application traces, and system availability with purpose-built UIs.



Security

Prevent, collect, detect, and respond to threats for unified protection across your infrastructure.



Analytics

Explore, visualize, and analyze your data using a powerful suite of analytical tools and applications.

Get started by adding integrations

To start working with your data, use one of our many ingest options. Collect data from an app or service, or upload a file. If you're not ready to use your own data, play with a sample data set.

[Add integrations](#)[Try sample data](#)[Upload a file](#)

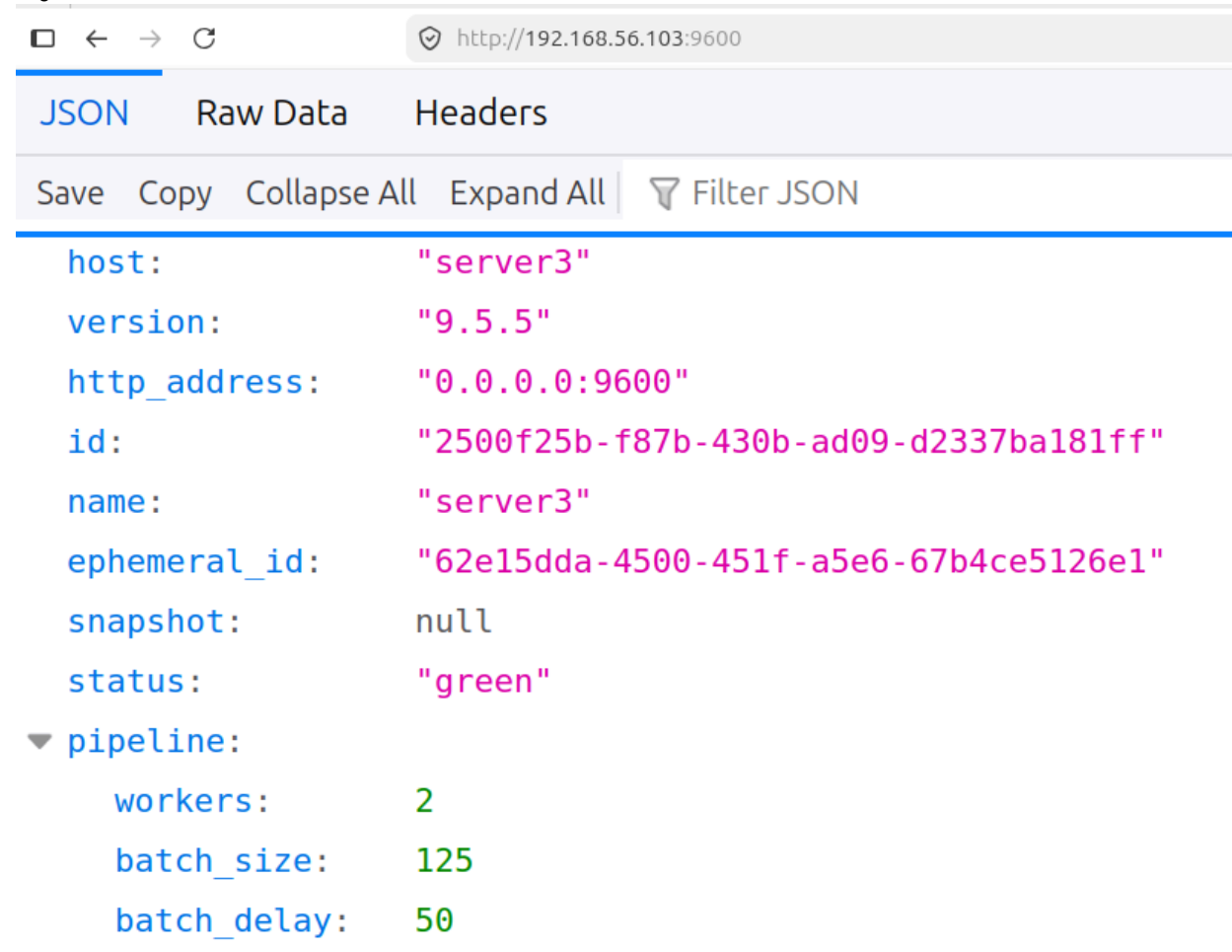


Cloud Connect

Use Elastic Cloud services like AutoOps and Elastic Inference Service for your self-managed clusters.

[Get started](#)

Logstash API



The screenshot shows a web browser displaying the Logstash API response. The address bar shows the URL `http://192.168.56.103:9600`. The response is in JSON format, with tabs for JSON, Raw Data, and Headers. The JSON is expanded, showing the following structure:

```

{
  "host": "server3",
  "version": "9.5.5",
  "http_address": "0.0.0.0:9600",
  "id": "2500f25b-f87b-430b-ad09-d2337ba181ff",
  "name": "server3",
  "ephemeral_id": "62e15dda-4500-451f-a5e6-67b4ce5126e1",
  "snapshot": null,
  "status": "green",
  "pipeline": {
    "workers": 2,
    "batch_size": 125,
    "batch_delay": 50
  }
}

```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

1. What are the benefits of having a log monitoring tool?

A log monitoring tool like the ELK stack centralizes logs from many hosts into one searchable place so you can spot errors, security events, and anomalies without SSHing into each machine individually. It also enables real-time alerting, trend analysis, and root-cause investigations that would be nearly impossible with scattered flat log files.

Conclusions:

I conclude that deploying the ELK stack through Ansible successfully automated the installation and configuration of Elasticsearch, Kibana, and Logstash across three separate virtual machines, with each component isolated to its own host to work within memory constraints. Despite challenges such as the X-Pack security conflict, I was able to determine that Ansible is an effective tool for managing reproducible, multi-node log monitoring infrastructure.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT



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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

**Purpose of
Use:** _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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